ENGINEERING TRIPOS PART IIA

ELECTRICAL AND INFORMATION ENGINEERING
TEACHING LABORATORY

MODULE EXPERIMENT 3F1

FLIGHT CONTROL

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Objectives:

- Simulation of various aircraft models on the computer.
- Study real-time (manual) control and the limitations imposed by time delays.
- Design of a simple autopilot.
- Illustrate frequency response concepts in analogue and digital control systems, conditions for oscillation in feedback systems and stability.
- Gain familiarity with Python and Jupyter Notebooks.

3F1 Lab Worksheet

2.1 Simplified Aircraft Model

Transfer function $= \dfrac{N}{s^2 + Ms} = \dfrac{10}{s^2 + 10s}$

`num = [10]`   `den = [1, 10, 0]`

2.2 Modelling Manual Control

Controller transfer function $= K(s) = ke^{-Ds}$

$k = 0.8$ $D = 1$ s

Phase margin = 45 degrees

Amount of extra time delay which can be tolerated $= \dfrac{45 \times \pi/180}{0.8} \approx 1.0$ s

2.3 Pilot Induced Oscillation

Period of oscillation (observed) = 5 s

Period of oscillation (theoretical) $= \dfrac{2\pi}{\omega_{pc}} = \dfrac{2\pi}{0.8} \approx 7.85$ s

2.4 Sinusoidal disturbances

Maximum stabilising gain = 4.5

Gain at 0.66 Hz = 5.5 dB   Phase at 0.66 Hz = −90 degrees

2.5 Unstable Aircraft

Fastest pole at $T = 0.6$

3.2 Autopilot with Proportional Control

Proportional gain $K_c = 12$ Period of oscillation $T_c = 2$ s

3.3 Autopilot with PID Control

Transfer function of PID controller $= K_p \left(1 + \dfrac{1}{T_i s} + T_d s\right)$

PID constants: $K_p = 7.2$ $T_i = 1.0$ $T_d = 0.25$

Adjusted value of $T_d = 0.35$

3.4 Integrator Wind-up

Integrator bound $Q = \dfrac{d \cdot T_i}{K_p} = \dfrac{2 \times 1.0}{7.2} = \dfrac{5}{18} \approx 0.278$

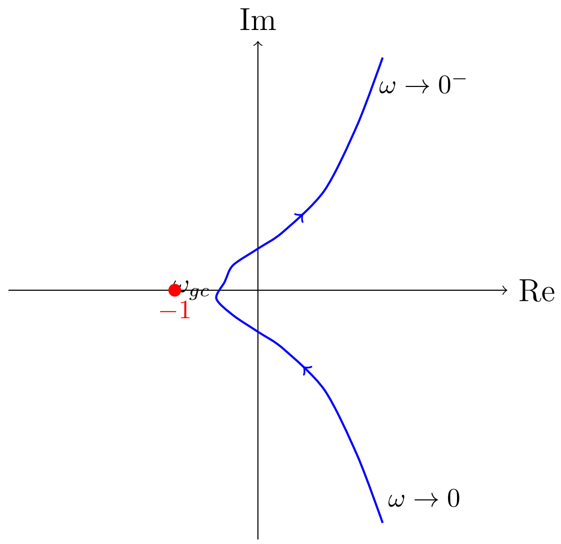
3F1 Flight Control Lab Report

1. (§2.2 Modelling Manual Control) Nyquist diagram (from Bode diagram) for controller in series with plant.

The open-loop transfer function is $K(s)G(s) = 0.8e^{-s} \cdot \dfrac{10}{s^2 + 10s}$.

From the Bode diagram (Figure 2):

- At low frequencies ($\omega \to 0$): magnitude $\to \infty$, phase $\to -90$
- At gain crossover ($\omega_{gc} = 0.8$ rad/s): magnitude $= 0$ dB, phase ≈ -135
- Phase crosses -180 at approximately $\omega = 1.5$ rad/s



Nyquist diagram of $K(s)G(s)$

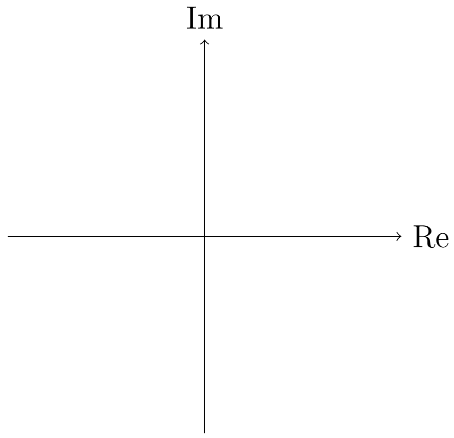
2. (§2.2 Modelling Manual Control) Are you using any integral action? Give a brief explanation. What does this imply about the accuracy of the model of the human controller?

3. (§2.3 Pilot Induced Oscillation) Explain the oscillation of the feedback loop. How does your observed period of oscillation compare to the theoretical prediction?

4. (§2.3 Pilot Induced Oscillation) Can you give a rough guideline to the control designer to make PIO less likely?

5. (§2.4 Sinusoidal Disturbances) Was your manual input able to reduce the error (as compared to providing no input)?

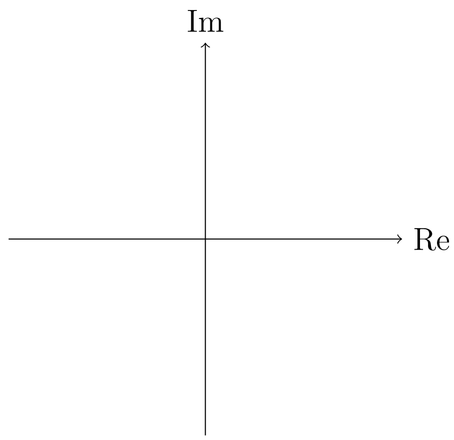
6. (§2.5 An Unstable Aircraft) Nyquist diagram for $G_2(s)$.



Nyquist diagram of $G_2(s)$

7. (§2.5 An Unstable Aircraft) Explain, using the Nyquist criterion, why the feedback system is stable with a proportional gain greater than 0.5.

8. (§2.5 An Unstable Aircraft) Sketch of a Nyquist diagram for $G_2(s)$. with a small time delay D.



Nyquist diagram with small time delay

9. (§3.4 Integrator Wind-up) Explain how you calculated the bound on Q.